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Change record

Issue	Revision Date	Section/Parag. affected	Reason/Initiation/Documents/Remarks
1	18/09/2026	all	First public release for pipeline version 4.6.4

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1 Introduction

1.1 Scope

This document describes how to reduce VISIR data with the `edps-gui` (Graphic User Interface), the dashboard of the ESO Data Processing System (EDPS), which is the recommended interface to reduce data from ESO telescopes. Details on the VISIR data reduction stream and how to configure the reduction to meet specific scientific needs are also given.

For a more extensive documentation on the `edps-gui` itself, consult the dedicated manual [here](#).

For a description of the VISIR pipeline itself, consult the pipeline manual available at: https://www.eso.org/sci/software/pipe_aem_table.html.

Note: this tutorial refers to:

- VISIR instrument pipeline named `visir`, version 4.6.4.
- VISIR workflow: `visir.visir_wkf`
- EDPS version 1.5.7.
- `edps-gui` version .91.

1.2 What is EDPS?

The ESO Data Processing System (EDPS) is a framework to run ESO's data processing pipelines and it is meant to eventually replace the previous [ESOReflex environment](#). The general principles of EDPS have been described by [Freudling, Zampieri, Coccato et al. \[2024, A&A, 681, A93\]](#). Please refer to that paper if you have used EDPS for research resulting in a scientific publication.

Each of ESO's data processing pipeline consists of a series of standalone programs called *recipes*. Each recipe is designed to process certain type(s) of input data. The processing of these input data typically requires a range of auxiliary files such as calibration files. EDPS is designed to select appropriate input data for the different recipes of a pipeline, and execute them in sequence. This is done by specifying for each pipeline the workflow for organizing data and executing the recipes. Then, the workflow can be used to process a set of data fully automatically.

1.3 Main concepts

EDPS is an environment designed to execute the recipes of an instrument pipeline according to a series of instructions. The main concepts in EDPS are:

- **Workflow and reduction cascades.** A workflow is a series of instructions designed to reduce data with an instrument pipeline in potentially multiple ways, by carrying on a sequence of tasks. Each workflow can define multiple reduction cascades, depending on the scientific needs. For example, the same workflow

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can be used to process data following different strategies that trigger different reduction steps (e.g. in one strategy flux calibration can be omitted) or different end-points (e.g., combine different science exposures, or stop after the reduction of individual exposures without combining them). Each of these "strategies" defines a "reduction cascade".

- **Task, jobs, and recipes.** A task is an element in the workflow that performs a given step of the data reduction cascade. Tasks are often associated to a recipe of the underlying instrument pipeline. A job is a work unit in a processing environment, that runs a recipe on a set of input data with a set of recipe parameters. A single task can generate several jobs: for example, a "bias" task, can generate multiple jobs, each of the running the bias recipe on a different set of input files.
- **Dataset.** A dataset is a collection of files, that are needed to perform the data reduction as specified by the workflow. It consists, for example, of one or more science files plus the calibrations needed to process them. In EDPS, datasets have a hierarchical structure, which highlights the connections between the various files and tasks (e.g., task A is an input to task B).
- **Target and Target category.** The "target", or the "target task" is the end point of the reduction cascade. When specifying a target, EDPS will process all and only the files needed to execute it. For example, if my target is "science", and the science files need the bias files, EDPS will process only the biases that have been selected to process those science files; then it processes the science using the product of the bias reduction. However, if my target is bias, then EDPS will process all and only the bias files, regardless they are not used by any science. In this case, EDPS does not process the science, as it has already reached the end reduction point (e.g., process all biases). The "Target category" is a group of targets that have similar purposes. For example, the target category "science", includes all the tasks that deliver final scientific products, the target category "qc1calib" includes all and only the tasks that processes calibrations (e.g., bias, flat fields, standard stars).

1.4 Installation

1.4.1 Prerequisites

Prerequisites for a well functioning installation of EDPS and EDPS-gui are:

- Recent Firefox or Chrome browser, Python 3.11 or higher (but there are issues with Python 3.14).
- At least one ESO pipeline with EDPS workflow should be in your system. To install the desired ESO pipelines, follow the instructions in the ESO pipelines pages. NOTE: the `apptainer` installation method is currently not supported. After the installation, the `esorex` command must be in the path. To test whether the installation was successful, type

```
esorex --recipes
```

A list of available recipes should appear.

- Install `graphviz`, `fv`, and `ds9`, which have to be included in the system path (defining aliases not enough). On linux, `Graphviz` can be easily installed via:

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```
sudo apt install graphviz (Debian, Ubuntu)
sudo dnf install graphviz (Fedora)
```

Check the [Graphviz](#) webpage for installation instructions for other OS.

`fv` and `ds9`, are optional. To install them, follow the instructions in corresponding webpages. You can test whether these three packages are installed and their paths are correctly set by typing on a terminal:

```
dot -V
fv -version
ds9 -version
```

1.4.2 Installation steps

To install EDPS follow these steps:

- Create a new Python virtual environment and activate it:

```
python3 -m venv edpsgui
. edpsgui/bin/activate
```

Make sure the `python3` version is 3.11 or higher, but not 3.14.

- Install the required packages:

```
pip install --extra-index-url \
https://ftp.eso.org/pub/dfs/pipelines/repositories/stable/src \
edps edpsgui edpsplot adari_core
```

To run the `edps-gui` type from a terminal (with the active environment):

```
edps-gui
```

Important note. The first time `edps-gui` is executed, you will be asked to specify the directory where the reduction products (fits files and quality plots) will be stored. The default location is `$HOME/EDPS_data`. During the first execution, a configuration file named `application.properties` will also be saved in the directory (newly created) `$HOME/.edps`.

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2 Reducing demo data

Follow this procedure to quickly reduce VISIR demo data. We assume that the EDPS, `edps-gui`, the VISIR pipeline and its associated demo data are installed in your system. For general instructions on how to install EDPS and the pipeline, see Section 1.4 or please visit: https://www.eso.org/sci/software/pipe_aem_main.html.

2.1 Setting the workflow

Proceed as follows:

1. If not done already, activate the EDPS virtual environment, defined during installation (Sect. 1.4).
2. Start the `edps-gui` dashboard by typing:

```
edps-gui
```

The `edps-gui` dashboard will start in a browser window (Figure 1).

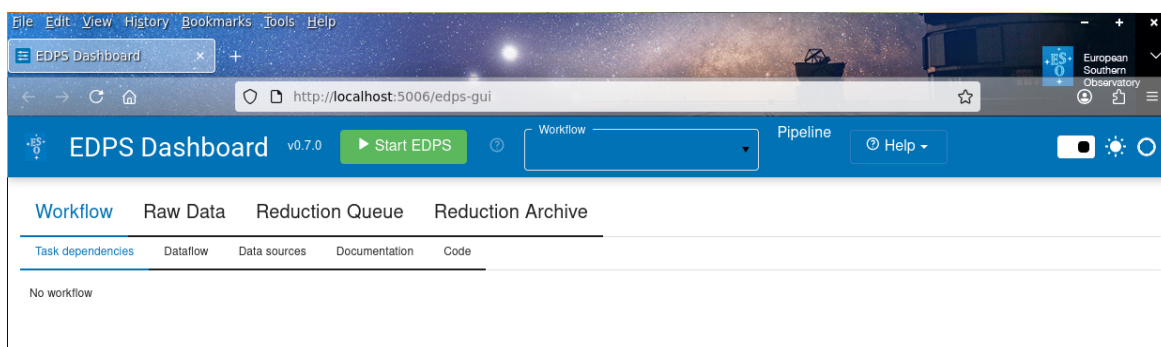


Figure 1: The empty `edps-gui` Dashboard; the underlying EDPS engine has not yet been started and no workflow has been loaded.

3. Optionally, before starting EDPS, one can specify new settings by pressing Help → Settings (Figure 2).

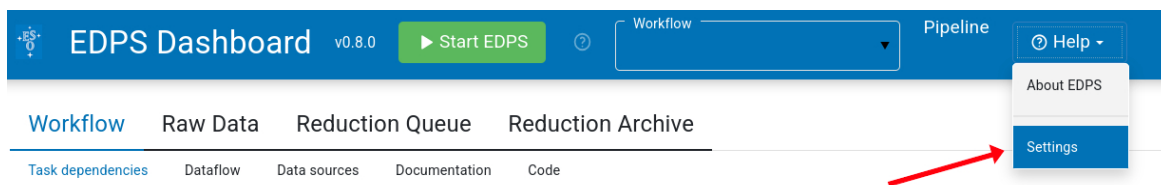


Figure 2: The “Help” → “Settings” menu.

4. On the browser window with the dashboard, press the button ‘Start EDPS’.
5. Choose the `visir.visir_wkf` workflow from the list in the ‘Workflow’ field. The workflows offered in this selector depend on the installed pipelines. The graphic workflow representation will appear as in Figure 4.

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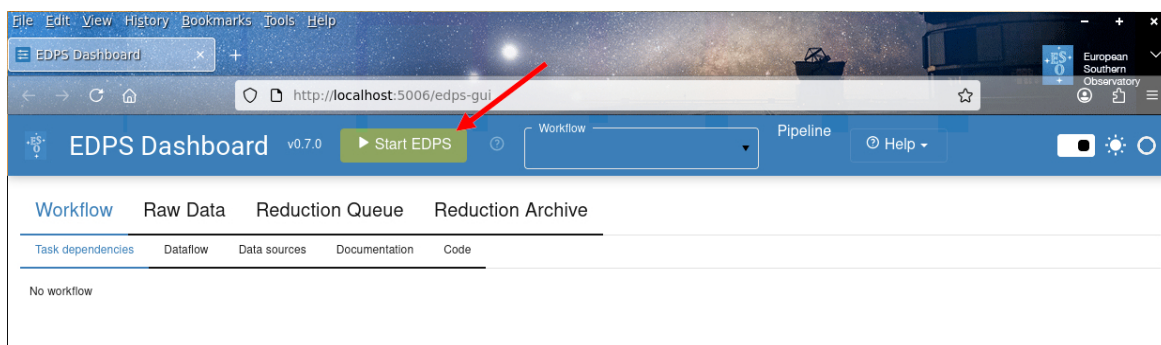


Figure 3: The “Start EDPS” button.

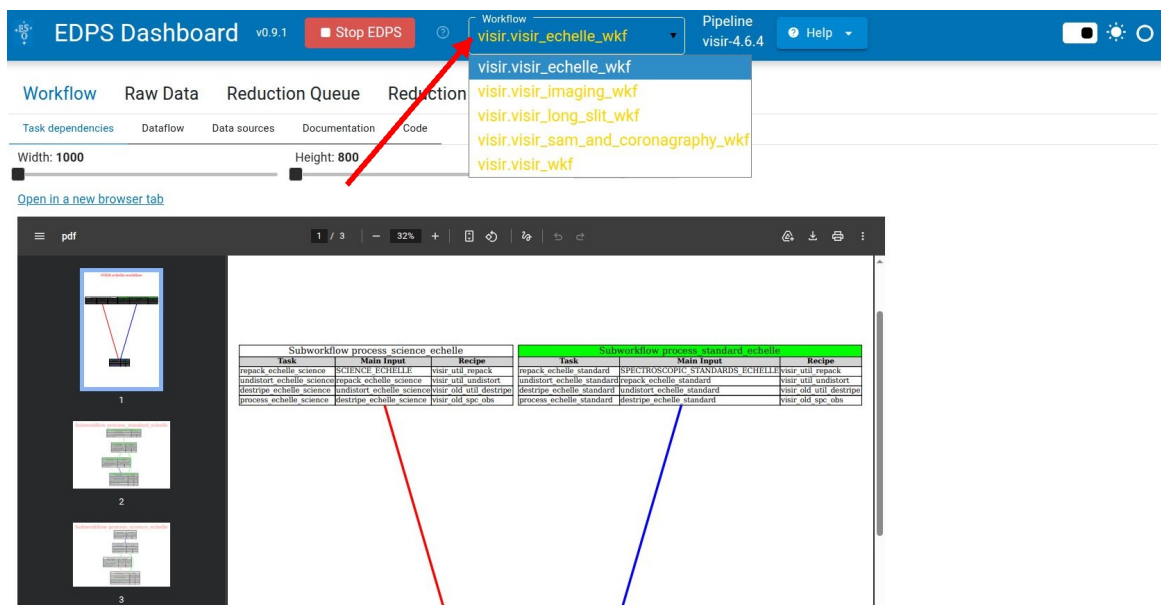


Figure 4: The edps-gui with the VISIR workflow loaded.

2.2 Selecting the input data

1. Press ‘Raw Data’ to enter the corresponding tab, as in Figure 5.

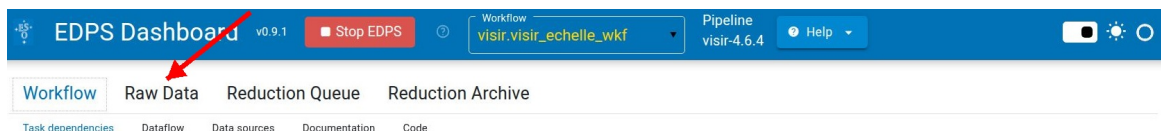


Figure 5: How to select RAW data Tab.

2. Press ‘Select Inputs’. A selection window will appear that allows to select data that are stored on a local disk (Figure 6).
3. (Optional). Select the reduction target, configure the workflow parameter and specify the association

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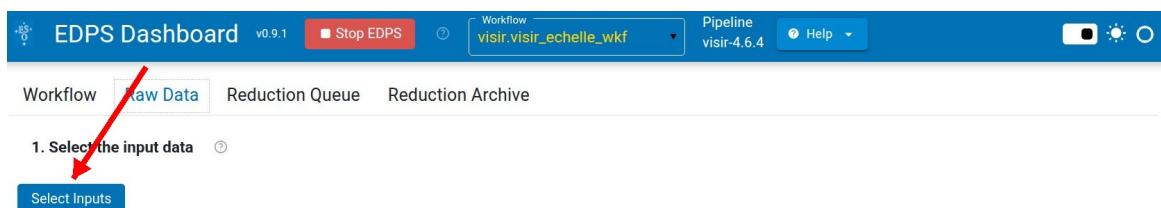


Figure 6: How to select input data.

preferences. These steps are optional. For more information see Section 4.

- Press 'Create Datasets'. A list of datasets appears, one line for each set of science data (Figure 7).

	Dataset	Reduction Target	Object	Exptime(s)	Files	CalibLevel	Complete	Submitted	Archived	ins.filt2.name	ins.grat
☐	VISIR.2009-06-26T03:14:04.354	calibrate_science_long_slit	IRAS-19114+0002	0.2	9	1	✓	✗	✗	N_SW	8.
☐	VISIR.2009-06-26T03:23:27.269	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_LW	11
☐	VISIR.2009-06-26T03:27:51.227	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_LW	12
☐	VISIR.2015-08-03T03:07:03.526	calibrate_science_long_slit	5526-A31-Tip-0.0-P42d-V10.967	0.1	15	1	✓	✗	✗	OPEN	Nr
☐	VISIR.2009-06-26T03:18:31.242	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_SW	9.

Figure 7: How to inspect the input data directory to create datasets.

- Choose the datasets that should be processed (Figure 8)

	Dataset	Reduction Target	Object	Exptime(s)	Files	CalibLevel	Complete	Submitted	Archived	ins.filt2.name	ins.grat
☒	VISIR.2009-06-26T03:14:04.354	calibrate_science_long_slit	IRAS-19114+0002	0.2	9	1	✓	✗	✗	N_SW	8.
☐	VISIR.2009-06-26T03:23:27.269	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_LW	11
☐	VISIR.2009-06-26T03:27:51.227	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_LW	12
☐	VISIR.2015-08-03T03:07:03.526	calibrate_science_long_slit	5526-A31-Tip-0.0-P42d-V10.967	0.1	15	1	✓	✗	✗	OPEN	Nr
☐	VISIR.2009-06-26T03:18:31.242	calibrate_science_long_slit	IRAS-19114+0002	0.1	9	1	✓	✗	✗	N_SW	9.

Figure 8: How to send the selected datasets to the Reduction Queue for processing.

and send them to the data reduction queue by pressing 'Submit to Reduction Queue'. Note that this action does not start the reduction automatically.

2.3 Start the reduction

- Press the 'Reduction Queue' tab (Figure 9).

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Figure 9: How to select Reduction Queue tab.

2. Select the datasets you'd like to reduce.
3. (Optional). Configure the workflow and recipe parameters by pressing the wheel button  to open the configuration editor. See Section 4.2 for more information on the configuration editor.
4. Press the 'Reduce' button (Figure 10). The selected data will now be processed with the configured parameters.

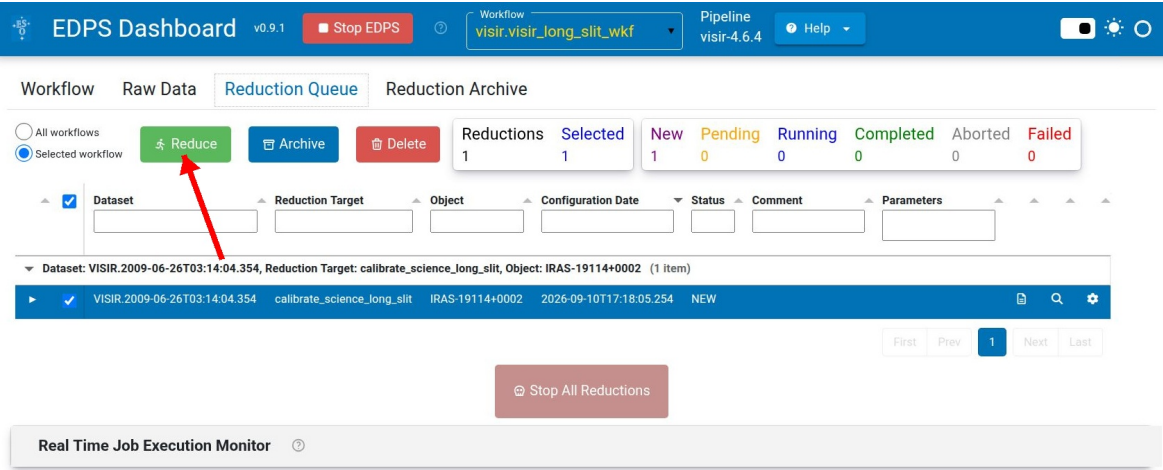


Figure 10: Reduce.

Congratulations! You reduced your first data with the EDPS dashboard! All the reduced data are saved in the EDPS_data directory specified when executing edps-gui for the first time.

2.3.1 Quality plots

It is possible to inspect the information on each job, such as quality plots showing the products (for the most important jobs and products), the list of inputs and output files, the recipe parameters and logs. All the information on the job processing can be inspected from the 'Reduction Queue' window. While quality plots are produced only for completed jobs, the other relevant information is available also for failed jobs.

The information associated for the main product can be inspected by pressing the magnifying glass symbol at the right side of each dataset. To inspect the information associated to other individual jobs (e.g., calibrations), proceed as follows:

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- Expand the desired dataset by pressing the black arrow on its left. The list of jobs will appear with the associated status (COMPLETED, RUNNING, PENDING, MISSING, ABORTED, FAILED)
- Press the magnifying glass symbol at the right side of the job you want to inspect.

Figure 11: How to look for job information from the Reduction Queue tab.

2.4 Exporting the final products

Completed reductions can be 'Archived' (i.e. declared 'completed' because no more work is needed) and removed from the Reduction Queue. Additionally, even if all products for all tasks are saved in the EDPS_data directory, the most important products can be 'exported' to a desired location.

To do so, proceed as follows:

1. In the 'Reduction Queue' tab, select the dataset and the dataset for which you want to export the final products, and press the 'Archive' button.
2. Go in the Reduction Archive tab and click on the 'Export' button. A new tab window appears where you can indicate the directory you want to copy your final products; finally press "Export" to copy the data.

Exported products are organized by 'DATASET' (named as the first scientific exposure of the dataset), and 'TIMESTAMP' (time of start of reduction)

The final products saved in the specified directory are:

- Imaging, Coronagraphy and Sparse aperture masking:

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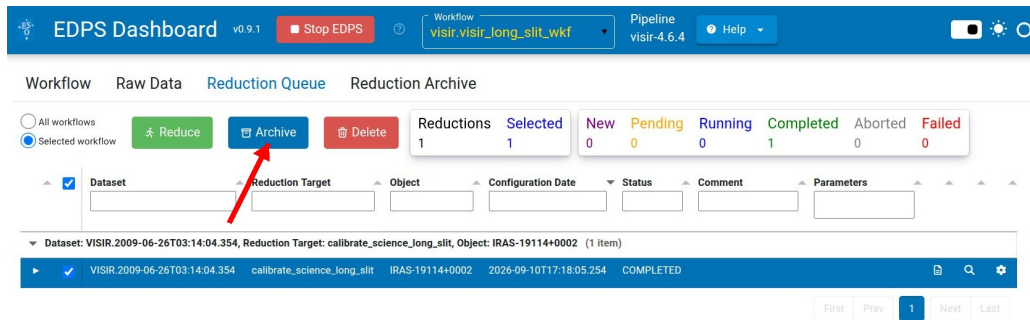


Figure 12: How to archive a completed reduction from the Reduction Queue tab.

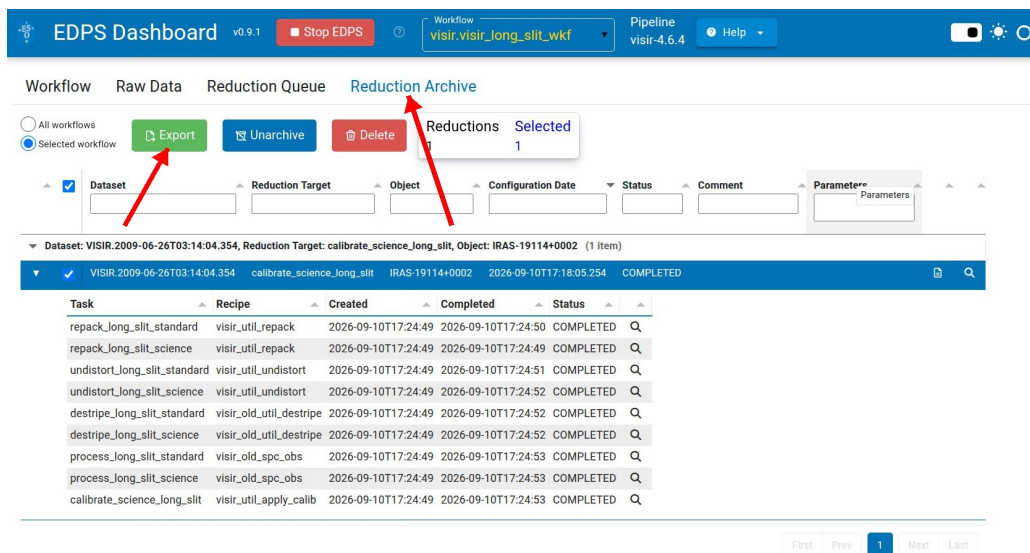


Figure 13: The reduction archive tab. This table contains all the different configurations of datasets that are declared "finished" and removed from the Reduction Queue. From this page, the user can export the most important files into a desired local directory.

- Multi-extension visir_util_join_bCOMBINED_001.fits file (PRO.CATG=IMG_OBJ_COMBINED or IMG_OBJ_COMBINED_BURST) that contains the following 2D extensions: flux, error, weights and a bad pixel mask (good pixels - 0 and bad pixels - 1). For more details, including a list of column names for the fits table, see the VISIR Pipeline User Manual, Sec. 9.4.1.
- All spectroscopic modes:
 - Multi-extension visir_spc_obs_tab.fits file (PRO.CATG=SPC_OBS_LMR_TAB or SPC_OBS_HRG_TAB) that contains for each order a 1D spectrum in a table and two 2D images with the flux and weights. If multiple orders are extracted, then these three extensions are replicated for each order. Finally, a 2D spectrum with all orders is also stored in a separate extension. For more details see the VISIR Pipeline User Manual, Sec. 9.4.2.

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Selected reductions

Dataset ▲	Configuration Date ▲	Status ▲	Parameters ▲
▼ Dataset: VISIR.2009-06-26T03:14:04.354 (1 item)			
VISIR.2009-06-26T03:14:04.354	2026-09-10T17:18:05.254	COMPLETED	

☒ Export all selected reductions
☐ Export the latest reduction for each selected dataset

Output directory

Export

Close

Figure 14: The EXPORT dialogue window, where the user can decide which reduced configuration to save and where.

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3 The VISIR data reduction flow.

The overall data flow of the VISIR pipeline is displayed in Figure 15.

The reduction cascade is organized in tasks, which represent well-defined steps in the process. Tasks can be grouped inside sub-workflows. Each task runs a recipe; the detailed description of the algorithms, input, outputs and recipe parameters used in each recipe are available in the pipeline manual. Here, we present only the description of most important features.

The `visir.visir_wkf` EDPS workflow is designed to execute the tasks that deliver the final reduced data cube for each dataset. It can be either the product of a single exposure, or the combination of multiple exposures. Only calibrations needed by the selected the scientific exposures are processed.

It is possible to set EDPS to perform the data reduction until a certain step of the reduction chain (e.g. to reduce only standar stars, or only flat fields). This is done by specifying the desired tasks in the field **Select reduction target** of the **Raw Data** tab.

The reduction steps of the `visir.visir_wkf` workflow are listed below. Before starting the reduction, the parameters of the recipes associated to each task can be configured by pressing the button  close to each dataset configuration. See for more info on the configuration editor 4.2

3.1 Imaging, Coronagraphy, Sparse aperture masking

- Data repacking and dark/sky subtraction (`visir_util_repack`). The input data is first split – the science images and, if present, the photometric standards are sent to be decoded, this step transforms the input FITS files into a format (still FITS) that makes the subsequent processing easier. The chop/nod background subtraction is applied.
- Detecting objects and calculating shifts (`visir_util_detect_shift`). This is critical for the subsequent image alignment. A sky background frame is created during this step, after applying a kappa-sigma clipping algorithm to the chop-nod corrected data, and finally the sky-subtracted images will be aligned with the appropriate offsets and co-added together to produce a final image.
- Shifting and adding individual frames together (`visir_util_run_swarp`).
- Optional photometric calibration that is carried out only if the input Data Set contains a photometric standard. Only the final combined image is calibrated, so if the users want to study the variability of a source during the observations, they must split the data into separate Data Sets, process them separately, and they will be calibrated individually. The stellar flux is corrected for the background, and the error budget includes the sky error. The aperture that maximizes the signal-to-noise ratio is used. The measured standard star flux is compared with the cataloged flux, in Jy. The final reported sensitivity is computed in units of mJy at $10\text{-}\sigma$ level in 1 hr (the conversion factor is the reciprocal of this value). If a conversion factor is derived for the Data Set, then the final fits are converted from ADU sec^{-1} into physical units Jy. For more details see the VISIR pipeline manual.

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3.2 Spectroscopy modes

- Data repacking (`visir_util_repack`): the same as the imaging case above.
- Distortion Correction (`visir_util_undistort`) is based on analytical parametrization. The parametrization was derived from an instrument model. The correction is applied directly by interpolating the distortion corrected pixel value from the source pixels. The pixels that are marked as bad are ignored during the interpolation.
- Wavelength Calibration and Spectral Extraction (`visir_util_detect_shift`, `visir_old_spc_wcal`, `visir_old_spc_wcal_ech`, `visir_old_spc_obs`, `visir_old_spc_obs_ech`), based on a polynomial that describes the dispersion relation. The VISIR physical model – represented by a first degree polynomial – is used as an initial guess. Experience has shown that only a linear shift is required and this is achieved by cross-correlating a sky spectrum derived from the data and a model of the atmospheric emission. The derived offset is then applied to the model. By default the pipeline extracts a 1D spectrum of the brightest source, applying the optimal extraction method.
- Photometric Calibration (`visir_old_spc_phot`, `visir_old_spc_phot_ech`) is carried out by comparing a flux model (available in the static calibration database) of the standard star with the observed flux. The final reported sensitivity is computed in units of mJy at the $10\text{-}\sigma$ level in 1 hr (the conversion factor is the reciprocal of this value). The normalization per unit time is obtained calculating the total integration time as follows: $\text{DIT} \times \text{NDIT} \times \text{NFILES} \times \text{NCHOP} \times 2$. The factor of two is due to the half-cycle chopping. For more details, see the VISIR pipeline manual.

For further details see VISIR Pipeline User Manual, Sec. 8. Note that the VISIR pipeline cascade omits two steps that the users might expect to see:

- separate dark subtraction step – because the dark signal is subtracted together with the sky emission;
- flat-fielding, – because the detector is too unstable, and the flat-fielding procedure would only add noise.

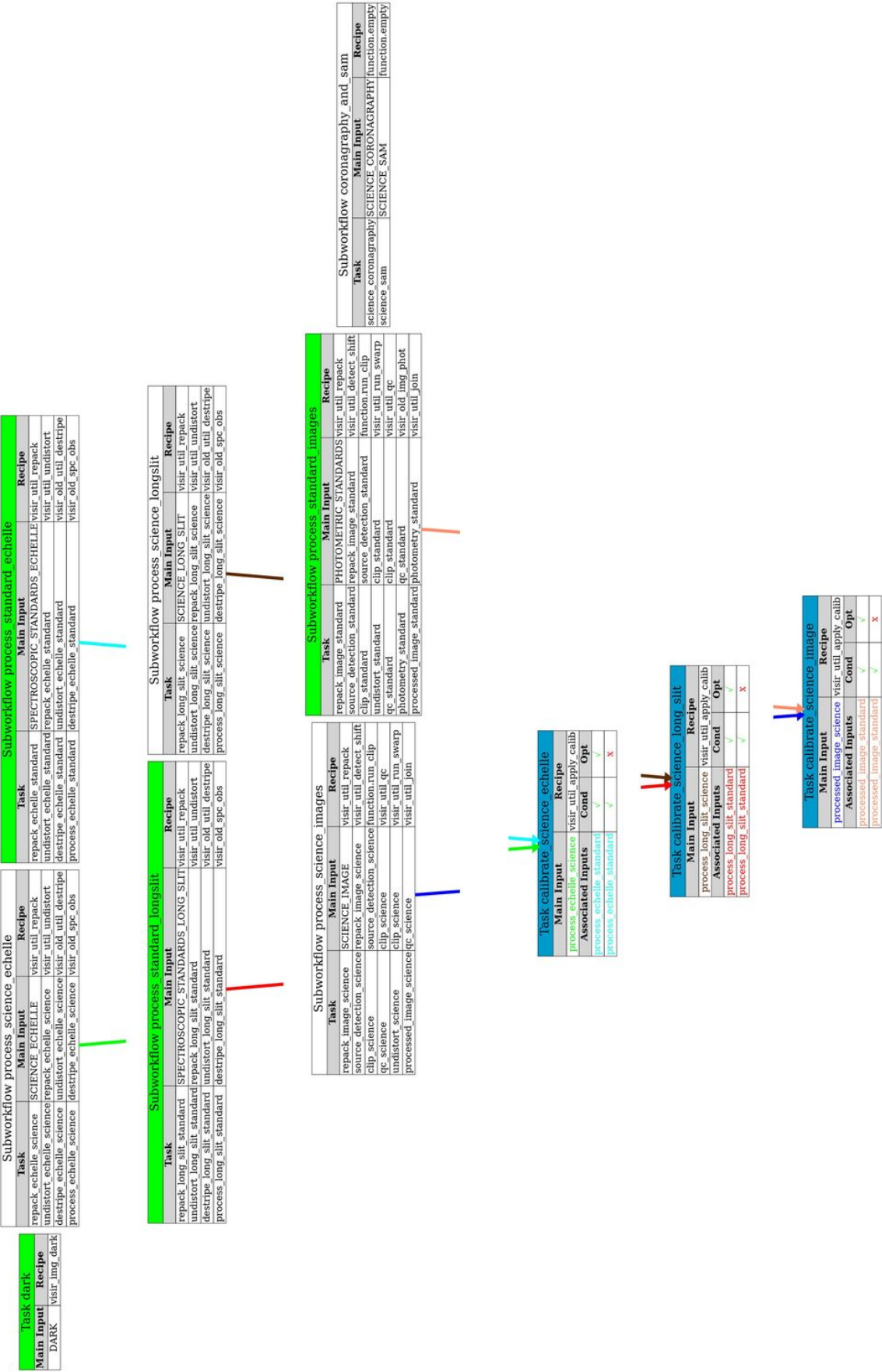


Figure 15: The data reduction cascade of the VISIR workflow.

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4 Overview of all the data reduction configuration options

4.1 Selection of most appropriate calibrations

By default, EDPS associates raw calibrations to the reduction process. It is also possible to use pre-processed calibrations (a.k.a. master calibrations) if available, in order to speed up the reduction. The preference can be specified in the Raw Data tab, before creating the datasets.

Possible values of the Calibration Preferences are:

- **raw_per_quality_level:** At equal quality of reduction, association of raw calibrations is preferred. This is the default.
- **master_per_quality_level:** At equal quality of reduction, association of master calibrations is preferred.
- **raw:** Association of raw calibration is preferred, despite the quality of results.
- **master:** Association of master calibration is preferred, despite the quality of results.

When master calibrations are used, the reduction step needed to process raw calibrations are not executed. The reduction then moves directly to the process of scientific exposures.

For example, if reduction speed for a quick check is preferred over a high quality reduction, one can select "master". In this case, old master calibrations are associated even if there are raw calibrations closer in time (and therefore more likely to ensure better quality products).

The quality level that the selected calibrations deliver is indicated close to each dataset in the 'Raw input' tab, under the column 'CalibLevel'. CalibLevel=0 indicates that calibrations that follow the rules of the instrument calibration plans have been selected. The higher the number, the poorer the quality of the products.

4.2 Configuration of parameters: the configuration editor

The data reduction of each dataset can be configured according to the scientific needs using an appropriate configuration editor.

The EDPS workflows contain two types of parameters and they both have default values that can be modified to improve the data reduction.

- **Workflow parameters** (for some workflows only) are global and they are applied to the entire workflow. They are accessible both in the 'Raw Data' tab, prior to the creation of a dataset, and in the 'Reduction Configuration' editor, in the 'Reduction queue' tab. Note: some workflow parameters were already configured before creating the dataset and sending it to the reduction queue. Here, they can be changed again. Please, note that the parameters have an effect only on the files that are already in the dataset. If one specifies a parameter that should include extra files in the dataset (e.g., the inclusion of more calibrations), files are not added and the reduction might fail. If you need to change a parameter that modifies the dataset content, please go back to the Raw data tab and create a new dataset.

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- **Recipe parameters** are specific to the individual recipes and can be configured per task. They are accessible in the ‘Reduction Configuration’ editor, in the ‘Reduction queue’ tab.

This editor allows to configure the data reduction for a given dataset by specifying workflow and recipe parameters.

Note: some workflow parameters were already configured before creating the dataset and sending it to the reduction queue. Here, they can be changed again. Please, note that the parameters have an effect only on the files that are already in the dataset. If one specifies a parameter that should include extra files in the dataset (e.g., the inclusion of more calibrations), files are not added and the reduction might fail. If you need to change a parameter that modifies the dataset content, please go back to the Raw data tab and create a new dataset.

To open the editor, click on the wheel button  next to the dataset you desire to configure the reduction for. A window with the configuration editor appears as shown Figure 16.

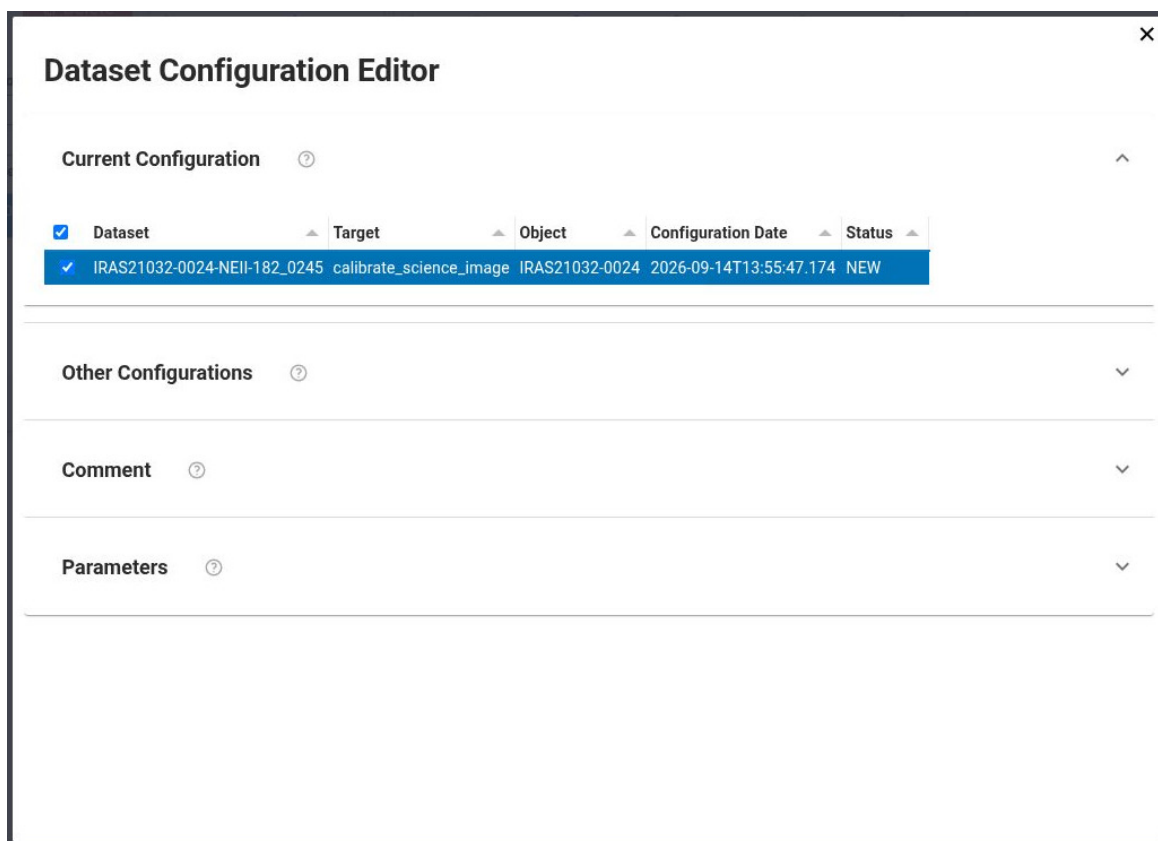


Figure 16: The Reduction Configuration editor. It contains 4 sections, that indicate the current configuration, list of other configurations to set, comments to insert, and the parameters to modify.

The editor is divided into 4 parts, which can be accessed pressing the corresponding expansion arrow.

- **Current configuration.** It indicates the name of the selected configuration for a given dataset (Figure 17).

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- **Other configurations.** It allows to specify other configurations, to which the changes shall be copied to (Figure 18).
- **Comment.** It allows to specify a comment to describe the configuration. It is possible to append or replace a comment (Figure 19). Comments can be changed on all configurations. It is possible to save the comment for the current configuration only, or for all the selected configurations.
- **Parameters.** A window as in Figure 20 appears.

The window allows to:

- Select the parameter set. A pre-determined list of workflow parameters and recipe parameters for a given use case. For the majority of the cases, the "science" parameter set can be used.
- Edit the workflow parameters. These are parameters that regulates the reduction strategy, e.g. whether to use a given calibration or not, or to trigger a certain reduction step. Note that if the changes imply that some files not in the dataset are needed, the reduction might fail. In case, go back to the raw data tab, edit the workflow parameters there, and recreate the datasets.
- Edit the recipe parameters. These are parameters associated to the recipe of a given task. Note: the same recipe parameters can be configured differently for the tasks that run the same recipe. Default parameters are shown (albeit some parameters can be dynamic, e.g. 'EDPS' changes their value depending on the type of input data).

Change the values according to the needs and then select whether to save it to the current or the selected configurations. Note, complete configurations cannot be modified, new configurations will be automatically created instead.



Figure 17: The first part of the Reduction Configuration Editor, that indicates the selected configuration.

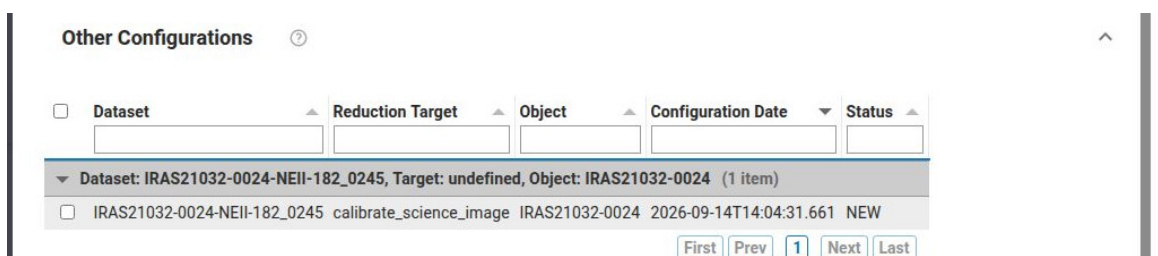


Figure 18: The second part of the Reduction Configuration Editor, that indicates other configurations for which we'd like to apply the changes.

OTHER CONFIGURATIONS

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Comment ?

Comment

This is a comment describing the reduction

☒ append
 ☐ replace ?

Figure 19: The third part of the Reduction Configuration Editor, that allows to specify a comment to the selected configurations.

Parameters ?

Parameter set
science_parameters

Workflow parameters

Parameter	Default value	Custom value
force_standard	FALSE	

Click on a parameter to view its description

Recipe parameters

Task
process_long_slit_science

Parameter	Default value	Custom value
visir.visir_old_spc_obs.nodding	NONE	
visir.visir_old_spc_obs.auto_bpm	TRUE	
visir.visir_old_spc_obs.rem_glitch	FALSE	
visir.visir_old_spc_obs.purge_bad	FALSE	
visir.visir_old_spc_obs.union	TRUE	
visir.visir_old_spc_obs.rej	0-0	
visir.visir_old_spc_obs.bkgcorrect	TRUE	

Figure 20: The fourth part of the Reduction Configuration Editor, that allows to specify the parameters sets and the recipe parameter per task. These settings can be applied to the "Selected Configuration" (Fig. 17) or to the "Other Configurations" (Fig. 18).

There are limited possibilities to improve the quality of the final product during the processing, and the improvement on one aspect may easily lead to a degradation in another. Perhaps, the most promising is to experiment with the spectral extraction parameters. The recipes `visir_spc_reduce` and `visir_old_spc_obs` accept an aperture file describing the options to extract spectra from multiple objects along the slit. The file name and path is defined in the parameter `-apfile`. This has to be defined in the parameter space of the workflow before running the workflow. The aperture file defines one extraction aperture per line, there can be many aper-

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tures/lines, each with its own extraction extraction method, sky region and sky measurement method. For a full description of the apfile see VISIR Pipeline User Manual, Sec. 9.3.2.

Other parameters to attempt adjusting are `-bkgcorrect` (for a column-based background subtraction) and `-destripe`.

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5 List of workflow tasks

This is the list of all the tasks and associated recipes in the VISIR workflow. Only some of them are needed for scientific reduction, they are indicated by the flag "yes" (triggered by default) or "optional" (triggered only if requested by a workflow parameter). Other tasks are not used for scientific reduction (they are indicated by the flag "no"), they are mainly used for instrument monitoring and they can be executed only by specifying them as target. Note that, when a task is specified as target, all the tasks that generate the calibrations needed for it are automatically executed.

TASK	RECIPE	Used in science reduction	Notes
calibrate_science_echelle	visir_util_apply_calib	no	
calibrate_science_image	visir_util_apply_calib		
calibrate_science_long_slit	visir_util_apply_calib		
dark	visir_img_dark		
destripe_echelle_science	visir_old_util_destripe		
destripe_echelle_standard	visir_old_util_destripe		
destripe_long_slit_science	visir_old_util_destripe		
destripe_long_slit_standard	visir_old_util_destripe		
photometry_standard	visir_old_img_phot		
process_echelle_science	visir_old_spc_obs		
process_echelle_standard	visir_old_spc_obs		
processed_image_science	visir_util_join		
processed_image_standard	visir_util_join		
process_long_slit_science	visir_old_spc_obs		
process_long_slit_standard	visir_old_spc_obs		
qc_science	visir_util_qc		
qc_standard	visir_util_qc		
repack_echelle_science	visir_util_repack		
repack_echelle_standard	visir_util_repack		
repack_image_science	visir_util_repack		
repack_image_standard	visir_util_repack		
repack_long_slit_science	visir_util_repack		
repack_long_slit_standard	visir_util_repack		
source_detection_science	visir_util_detect_shift		
source_detection_standard	visir_util_detect_shift		
undistort_echelle_science	visir_util_undistort		
undistort_echelle_standard	visir_util_undistort		
undistort_long_slit_science	visir_util_undistort		
undistort_long_slit_standard	visir_util_undistort		
undistort_science	visir_util_run_swarp		
undistort_standard	visir_util_run_swarp		

Table 5.0.0: VISIR pipeline tasks overview

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6 Frequently Asked Questions

- **Q1) Where can I find the final reduced data?**

Answer: all the products of all the datasets and the reductions are saved into the EDPS_data directory, specified when executing the edps-gui for the first time. One can decide to export only the final products for selected datasets and only for the desired reduction attempts into another location for further analysis. See Section 2.4 for further instructions.

- **Q2) How do I stop the application?**

Answer: Proceed as follows:

1. Press “Stop EDPS” in the Dashboard.
2. Type Ctrl-C in the terminal where the application is running. If the application doesn’t terminate, type Ctrl-C again.
3. Alternatively, kill the ‘panel serve’ process on your system, for example:

```
ps -e | grep panel # get the process ID of the gui (<pid>).
kill -9 <pid>
```

- **Q3) I have closed the browser window where the application is running. How can I reopen the application?**

Answer: Point your browser to: `http://localhost:5006/edps-gui`

- **Q4) Where can I find some data that I can use to test the application?**

Answer: Install the ‘datademo’ package provided with the pipeline installation or download the “Demo Data” package from https://www.eso.org/sci/software/pipe_aem_table.html.

Please note that the demo data can be large (tens of Gigabytes).

A convenient script to download demo data for any pipeline is also available and can be used from the command line:

```
curl -O https://eso.org/sci/software/apptainer/eso_download_demodata.sh
bash ./eso_download_demodata.sh
```

- **Q5) How can I start the edps-gui if the following message appears?**

```
Cannot start Bokeh server, port 5006 is already in use
```

Answer: The panel server was not closed properly. Kill it by typing:

```
ps -e | grep panel # get the process ID of the gui (<pid>).
kill -9 <pid>
```

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• **Q6) How do I get additional support on EDPS or data reduction in general?**

Answer: For suggestions, questions, or feedback in general, please open a ticket with the EDPS Support team. This https://support.eso.org/new-ticket?ticket%5Bticket_field_13%5D%5Bdata%5D=227 should take you directly to a webpage for creating and EDPS feedback ticket, but in case you want to navigate there 'manually', go to <https://support.eso.org>, login, click on "Submit Helpdesk Ticket", and specify the Help topic: "Post Observations", "ESO Data Processing System [EDPS]".

• **Q7) I have a lot of disk space, but when I install EDPS with pip or an ESO pipeline with Homebrew I get the error message: Cannot mkdir: No space left on device. How do I fix it?**

Answer: This depends on how much disk space is allocated to the /home, /var, and /tmp directories. The final solution would be to resize the space allocated to the in the organization of the filesystem. However, we list here few tricks that might do the job.

- Clearing the pip .cache to make space for new packages. Type the command:

```
pip cache purge
```

before installing EDPS.

- Redirect the cache, Homebrew temporary build directories into a partition with enough space. Set some of the following environmental variables in your .bashrc file:

```
export HOMEBREW_CACHE=<path_to_new_cache_directory>
export XDG_CACHE_HOME=<path_to_new_cache_directory>
export HOMEBREW_TEMP=<path_to_new_temporary_directory>
export TMPDIR=<path_to_new_temporary_directory>
```

The first moves only the location of Homebrew cache, the second the cache of most applications (instead of the default /home/username/.cache), the third moves the directory where Homebrew builds, extracts, and saves temporary files (instead of the defaults /tmp and /var/tmp). The last changes the global system temporary directory and affects most of the linux commands.

- As extreme measure, one can move the /home/linuxbrew/.linuxbrew directory somewhere else, and create a symbolic link in /home/linuxbrew. For example:

```
cd /home/linuxbrew
mv -f .linuxbrew <path_to_new_directory>
ln -s <path_to_new_directory> .linuxbrew
```

Important note: this operation might break some internal links. Recipes requiring external packages such as telluriccorr might not work (impacts on KMOS, XSHOOTER, FORS2, and MOLECFIT pipelines).

• **Q8) How can I manually run the recipes executed by EDPS?**

Answer: The directory ~/EDPS_data/VISIR/ contains individual subdirectories for each recipe, e.g., calibrate_science_echelle, calibrate_science_image, calibrate_science_long_slit, clip_science, clip_standard, destripe_echelle_science, etc. – one for each recipe. They contain randomly named sub-directories for each execution of the particular recipe, e.g., 0ea47c20-26dd-4ffd-8de2-9cad4bbf515a,

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2a73f7e3-aa4f-4853-baff-33de29c4395e, etc.

The users can find in those executable scripts names `cmdline.sh`, together with the necessary parameter files `*.rc` and lists of input fits files `*.sof`. To manually start the execution type at the prompt:

```
./cmdline.sh
```

It is advisable to make copies of all files before making changes.

- **Q9) Are the VISIR spectral products corrected for telluric absorption?**

Answer: Yes, during “normal” data processing the telluric correction is applied dividing the science spectrum by a telluric spectrum. The users can alternatively apply the `Molecfit` tool¹ outside the VISIR pipeline.

- **Q10) If the data set contains a spectrophotometric standard, and there are multiple apertures applied to the standard, then which apertures are used to correct the science target?**

Answer: As this opens up too many possibilities, we suggest keeping it simple and always using the aperture from layer 1 of the standard. This approach is robust. The users are free to define additional apertures on the standard, e.g. for data quality purposes, but the first one is always used. Also, the default values of `rl` and `rr` for this `visir_spc` workflow should be set to 75 to avoid the negative images.

¹<https://www.eso.org/sci/software/pipelines/>